Student Attendance Monitoring System with QR Code Integration for Higher Education Institutions

A Thesis Project presented to the

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In partial fulfillment of the requirements for the degree of

Bachelor of Science in Information Technology

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ABSTRACT

Title: Student Attendance Monitoring System with QR Code Integration for Higher Education Institutions

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Degree: Bachelor of Science in Information Technology

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Project Description:

This study presents the development of a web-based Student Attendance Monitoring System that utilizes QR code technology to automate and improve the process of recording student attendance in higher education institutions. Traditional attendance methods, such as manual roll calls and paper-based logs, are often time-consuming, prone to errors, and vulnerable to manipulation.

The proposed system allows instructors to generate unique QR codes for each class session, which students can scan using their mobile devices to mark their attendance. The system records real-time attendance data, ensuring accuracy and

minimizing fraudulent entries. It also provides features such as attendance reports, analytics, and notifications for absenteeism.

The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Results from user testing indicate that the system improves efficiency, reduces manual workload, and enhances transparency in attendance tracking. The system offers a reliable and scalable solution for modern academic environments.

CHAPTER 1: INTRODUCTION

I. Introduction

Monitoring student attendance is an essential task in educational institutions, as it directly impacts academic performance and student accountability. However, traditional methods such as manual attendance sheets and roll calls are inefficient, time-consuming, and susceptible to human error and dishonesty.

With the advancement of digital technologies, there is a growing need for automated systems that can streamline attendance tracking. QR code technology provides a fast, contactless, and reliable way to record attendance while minimizing manual intervention. This study proposes a web-based Student Attendance Monitoring System with QR code integration that enables efficient attendance recording, real-time monitoring, and automated reporting for higher education institutions.

II. Significance of the Study

- This system will benefit the following:
- Students – Faster and more convenient attendance logging
- Faculty – Reduced administrative workload and accurate records
- Administrators – Centralized monitoring of attendance data
- Institutions – Improved transparency and efficiency in attendance management

III. Statement of the Problem

General Problem

How can a QR code-based system improve the efficiency and accuracy of student attendance monitoring?

Specific Problems

1. How to automate attendance recording to reduce manual effort?
2. How to prevent fraudulent attendance entries (e.g., proxy attendance)?
3. How to provide real-time attendance tracking and reporting?
4. How to securely store and manage attendance data?

IV. Scope and Limitations

Scope

- QR code-based attendance logging
- Web-based dashboard for monitoring
- User roles: Admin, Instructor, Student
- Attendance reports and analytics

Limitations

- Requires internet connection for real-time functionality
- Depends on student access to mobile devices

- Limited to institutional use (not for external training programs)
- No AI-based recommendations (for now)

OPERATIONAL DEFINITION OF TERMS

Administrator – The personnel responsible for managing system users, classes, and reports.

Attendance Monitoring – The process of recording and tracking student presence in scheduled classes.

Authentication – The process of verifying user identity before granting access to the system.

Authorization – The process of assigning user roles and permissions.
Backend – The server-side component responsible for processing attendance data and managing the database.

Database – A structured collection of data that stores attendance records, user information, and class details.

Frontend – The user interface where students and instructors interact with the system.

QR Code (Quick Response Code) – A machine-readable code that stores data and can be scanned using a mobile device to record attendance.

Real-Time Processing – Immediate recording and updating of attendance data as it is scanned.

Report Generation – The feature that produces summaries and analytics of attendance records.

Role-Based Access Control (RBAC) – A system that restricts access based on user roles (Admin, Instructor, Student).

Session – A specific class schedule during which attendance is recorded.

System User – Any individual interacting with the system.

Usability (ISO 25010) – The ease with which users can use the system effectively.

Security (ISO 25010) – Measures implemented to protect attendance data from unauthorized access.